

12.55

EE25BTECH11042 - Nipun Dasari

Question:

For a matrix

$$\mathbf{M} = \begin{pmatrix} \frac{4}{5} & \frac{3}{5} \\ \frac{3}{5} & x \end{pmatrix} \quad (0.1)$$

The transpose of the matrix is equal to inverse of the matrix, i.e.,

$$\mathbf{M}^T = \mathbf{M}^{-1} \quad (0.2)$$

The value of x is given by

Solution: From (0.2) it is evident that the matrix (0.1) is an orthogonal matrix.

We can proceed by the fact that columns of an orthogonal matrix are pairwise perpendicular

$$\mathbf{M} = (\mathbf{c}_1 \quad \mathbf{c}_2) \quad (0.3)$$

where

$$\mathbf{c}_1 = \begin{pmatrix} \frac{4}{5} \\ \frac{3}{5} \end{pmatrix}, \quad \mathbf{c}_2 = \begin{pmatrix} \frac{3}{5} \\ x \end{pmatrix} \quad (0.4)$$

By using (0.2):

$$\mathbf{c}_1^T \mathbf{c}_2 = 0 \quad (0.5)$$

$$\Rightarrow \begin{pmatrix} \frac{4}{5} & \frac{3}{5} \end{pmatrix} \begin{pmatrix} \frac{3}{5} \\ x \end{pmatrix} = 0 \quad (0.6)$$

$$\Rightarrow \frac{12}{25} + \frac{3x}{5} = 0 \quad (0.7)$$

$$\Rightarrow x = -\frac{4}{5} \quad (0.8)$$